

Advanced Mathematical Techniques

Bookwork 9

1. **Problem 8.1.23:** The probability density associated with the normal distribution of statistics is given by:

$$f(x) = \frac{1}{\sigma(2\pi)^{1/2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

with $(-\infty, \infty)$ for the range of x . Show that

- (a) the mean value of x , $\langle x \rangle$ is equal to μ .

$$\begin{aligned} \langle x \rangle &= \int_{-\infty}^{\infty} \frac{x}{\sigma(2\pi)^{1/2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx \xrightarrow{z=\frac{x-\mu}{\sigma\sqrt{2}}} \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} (z\sigma\sqrt{2} + \mu) e^{-z^2} dz \\ &= \frac{\sigma\sqrt{2}}{\sqrt{\pi}} \int_{-\infty}^{\infty} z e^{-z^2} dz + \frac{\mu}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-z^2} dz = 0 + \mu = \mu \quad \checkmark \end{aligned}$$

- (b) the standard deviation $(\langle x^2 \rangle - \langle x \rangle^2)^{1/2}$ is given by σ .

$$\begin{aligned} \langle x \rangle^2 &= \mu^2 \\ \langle x^2 \rangle &= \int_{-\infty}^{\infty} \frac{x^2}{\sigma(2\pi)^{1/2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx \xrightarrow{z=\frac{x-\mu}{\sigma\sqrt{2}}} \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} (z\sigma\sqrt{2} + \mu)^2 e^{-z^2} dz \\ &= \frac{2\sigma^2}{\sqrt{\pi}} \int_{-\infty}^{\infty} z^2 e^{-z^2} dz + \frac{2\mu\sigma\sqrt{2}}{\sqrt{\pi}} \int_{-\infty}^{\infty} z e^{-z^2} dz + \frac{\mu^2}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-z^2} dz = \sigma^2 + 0 + \mu^2 = \mu^2 + \sigma^2 \\ \therefore \sqrt{\langle x^2 \rangle - \langle x \rangle^2} &= \sqrt{\mu^2 + \sigma^2 - \mu^2} = \sqrt{\sigma^2} = \sigma \quad \checkmark \end{aligned}$$

2. **Problem 8.1.24:** From the gamma distribution

$$f(x) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-\frac{x}{\beta}} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

show that:

- (a) $\langle x \rangle$ (mean) = $\alpha\beta$

$$\langle x \rangle = \int_{-\infty}^0 0 \cdot x dx + \int_0^{\infty} \frac{x}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-\frac{x}{\beta}} dx = \frac{1}{\beta^\alpha \Gamma(\alpha)} \int_0^{\infty} x^\alpha e^{-\frac{x}{\beta}} dx \xrightarrow{u=\frac{x}{\beta}} \frac{\beta}{\Gamma(\alpha)} \int_0^{\infty} u^\alpha e^{-u} du = \alpha\beta \quad \checkmark$$

- (b) σ^2 (variance) $\equiv \langle x^2 \rangle - \langle x \rangle^2 = \alpha\beta^2$

$$\begin{aligned} \langle x \rangle^2 &= \alpha^2 \beta^2 \\ \langle x^2 \rangle &= \int_0^{\infty} \frac{x^2}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-\frac{x}{\beta}} dx = \frac{\beta^2}{\Gamma(\alpha)} \int_0^{\infty} u^{\alpha+1} e^{-u} du = \beta^2(\alpha+1)\alpha \\ \therefore \langle x^2 \rangle - \langle x \rangle^2 &= \alpha^2 \beta^2 + \alpha\beta^2 - \alpha^2 \beta^2 = \alpha\beta^2 \quad \checkmark \end{aligned}$$

3. The beta distribution of probability theory has a probability density:

$$f(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$$

with x restricted to the interval $(0, 1)$. Show that:

$$(a) \langle x \rangle \text{ (mean)} = \frac{\alpha}{\alpha + \beta}$$

$$\frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 x \cdot x^{\alpha-1} (1-x)^{\beta-1} = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 x^\alpha (1-x)^{\beta-1} = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{\Gamma(\alpha + 1)\Gamma(\beta)}{\Gamma(\alpha + \beta + 1)} = \frac{\alpha}{\alpha + \beta} \quad \checkmark$$

$$(b) \sigma^2 \text{ (variance)} \equiv \langle x^2 \rangle - \langle x \rangle^2 = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

$$\langle x \rangle^2 = \frac{\alpha^2}{(\alpha + \beta)^2}$$

$$\langle x^2 \rangle = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 x^{\alpha+1} (1-x)^{\beta-1} dx = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{\Gamma(\alpha + 2)\Gamma(\beta)}{\Gamma(\alpha + \beta + 2)} = \frac{(\alpha + 1)\alpha}{(\alpha + \beta + 1)(\alpha + \beta)}$$

$$\begin{aligned} \therefore \langle x^2 \rangle - \langle x \rangle^2 &= \frac{(\alpha + 1)\alpha}{(\alpha + \beta + 1)(\alpha + \beta)} - \frac{\alpha^2}{(\alpha + \beta)^2} = \frac{\alpha^3 + \alpha^2 + \alpha^2\beta + \alpha\beta - \alpha^3 - \alpha^2\beta - \alpha^2}{(\alpha + \beta + 1)(\alpha + \beta)^2} \\ &= \frac{\alpha\beta}{(\alpha + \beta + 1)(\alpha + \beta)^2} \quad \checkmark \end{aligned}$$